MOU PAUL

ML infrastructure engineer building data and compute systems for frontier-model pre- and post-training. Architected petabyte-scale pipelines processing trillions of tokens and reliability-critical container platforms supporting RL workloads.

ppymou@gmail.com
paul.mou.dev
github.com/moomou

Experience

Member of Technical Staff @ Microsoft AI, Redmond — Dec 2024-Present

- Architected a phased redesign of RL post-training sandbox infrastructure with researchers, pairing immediate capacity expansion with EROFS-to-OCI conversion to migrate 80% of rollout traffic onto a global container platform spanning 30+ clusters
- Architected petabyte-scale Spark/Ray pipelines processing trillions of tokens from 3B+ PDFs for LLM pre-training; ran ablations to refine data-mixture strategies
- Engineered embedding- and LLM-as-a-Judge-based filtering and deduplication workflows to curate high-fidelity pre-training datasets

Software Engineer @ Roblox, San Mateo — May 2022-Dec 2024

- Led the evolution of analytics ingestion infrastructure to trillions of events per day, delivering $10M+ in annual cloud savings through data lake schema standardization
- Standardized observability and operations through dashboards and alerts as code, runbooks, capacity planning, and incident postmortems

Software Engineer @ Twitter, San Francisco — Jun 2018-Apr 2022

- Scaled Smyte's anti-spam infrastructure 10x after acquisition through re-architecture and integration with Twitter's platform
- Hardened GDPR compliance by designing a system to anonymize personal data across anti-spam infrastructure

Software Engineer @ Smyte, San Francisco — Nov 2015-Jun 2018

- Core member of a four-engineer team that built and scaled anti-spam infrastructure protecting hundreds of millions of users
- Operated thousands of Kubernetes pods across multiple clusters and helped build Smyte's first ML stack for abusive-image detection

Selected Projects

Tiny World Model - Built a VAE/MDN-RNN world model and trained a GRPO controller entirely on imagined rollouts, improving held-out next-state R-squared from 0.04 to 0.906 and real-game survival from 47.6 to 262.4 steps

Flow Matching from Scratch - Built and trained a 114M-parameter flow-matching transformer, generating 256x256 images after 100 epochs in about 6 hours on one RTX 5090

Technical Skills

Python, Go, Rust; PyTorch, Ray, Spark; Kubernetes, Docker, Kafka, gRPC; distributed systems, ML data and compute infrastructure

Education

University of Waterloo, Bachelor's in Systems Design Engineering
Stanford University, Computer Science, Master (withdrew)